

SAT Math Mastery · Week 6A

Percents — Answer Key & Worked Solutions

Time 120 minutes: 90 min solving + 30 min self-check and error log

Topics percent change · chained percent changes · reverse percent · percent of a percent
· table subsets · markup & discount · percent vs. percentage points · mixtures
& concentrations · sample-to-population estimates

Name / Date _____

1 · SKILLS & IDEAS

- S1.** Compute a percent increase or decrease from old and new values, always dividing by the *old* value.
- S2.** Turn a percent statement into a multiplier ($1 \pm \frac{p}{100}$) and chain several multipliers.
- S3.** Reverse a percent change: find the original amount by dividing by the multiplier.
- S4.** Translate phrasing: “ $p\%$ of”, “ $p\%$ greater than”, “ $p\%$ less than”, “how many times”, “percent of a percent”.
- S5.** Choose the correct denominator in a table: the stated subgroup, not the grand total.
- S6.** Markup, discount, tax and profit problems, including layered adjustments.
- S7.** Separate a percent change from a change in percentage points.
- S8.** Mixtures and concentrations: track the amount of the pure substance, which is additive.
- S9.** Scale a sample proportion up to a population (and use complements).

2 · REQUIRED FACTS

- $p\%$ means $\frac{p}{100}$. “ A is $p\%$ of B ” means $A = \frac{p}{100}B$.
- “ $p\%$ greater than B ” means $B \left(1 + \frac{p}{100}\right)$; “ $p\%$ less than B ” means $B \left(1 - \frac{p}{100}\right)$.
- An increase of $p\%$ followed by a decrease of $p\%$ does *not* return to the start: $(1 + r)(1 - r) = 1 - r^2 < 1$.
- A percent change is measured against the starting (old) value; a change in two percentages is measured in percentage points.
- In a mixture, amounts of the pure substance add; the percentages themselves do not.
- In a two-way table, the denominator is the total of the group named after “of the”.

3 · REQUIRED FORMULAS

$$\begin{aligned} \text{percent change} &= \frac{\text{new} - \text{old}}{\text{old}} \times 100, & \text{new} &= \text{old} \left(1 \pm \frac{p}{100}\right), & \text{old} &= \frac{\text{new}}{1 \pm \frac{p}{100}} \\ \text{chain: } N(1 + r_1)(1 + r_2) \cdots, & \text{percentage-point change} &= p_2 - p_1, & \text{relative change} &= \frac{p_2 - p_1}{p_1} \\ \text{mixture: } c_1V_1 + c_2V_2 &= c_{\text{final}}(V_1 + V_2), & \text{population estimate} &= \frac{\text{sample count}}{\text{sample size}} \times N \end{aligned}$$

4 · MINI THEORY SHEET

Multiplier method. Write every change as a factor: $+25\% \rightarrow 1.25$, $-30\% \rightarrow 0.70$, $+8\%$ tax $\rightarrow 1.08$. Multiply factors for successive changes; divide by a factor to undo it. The net percent change is (product of factors) $- 1$.

Where each percent applies.

Wording	Meaning
A is $p\%$ of B	$A = \frac{p}{100} B$
A is $p\%$ greater than B	$A = \left(1 + \frac{p}{100}\right) B$
A is $p\%$ less than B	$A = \left(1 - \frac{p}{100}\right) B$
A is how many times B	$\frac{A}{B}$ (ratio of the two factors)
$p\%$ of $q\%$ of N	$\frac{p}{100} \cdot \frac{q}{100} N$

Markup and discount. Cost $\rightarrow$ list price (markup factor) $\rightarrow$ selling price (discount factor, applied to the list price). Profit = selling price – cost.

Percent vs. percentage points. Going from 8% to 6% is a drop of 2 percentage points and a drop of $\frac{2}{8} = 25\%$ in the rate itself.

Table subsets. Underline the group after “of” (the denominator) and the group described after “who” or “that” (the numerator).

Mixtures. Pure amount = concentration $\times$ volume. Add the pure amounts and the volumes separately, then divide.

Samples. Assume the sample proportion holds for the population: multiply the proportion by the population size.

Speed shortcuts.

Situation	Shortcut
up $r\%$ then down $r\%$	net factor $1 - r^2$ (e.g. 20%: 0.96, down 4%)
undo a +25%	divide by 1.25, which is a 20% decrease, not 25%
percent of B is A : find B	$B = \frac{A}{\text{the percent as a decimal}}$
a ratio of two percent-changes	divide the two multipliers
mixture	write one equation for the pure substance, one for volume
price = rate $\times$ quantity	multipliers multiply: $1.25 \times 0.72 = 0.90$

5 · GUIDED EXAMPLES

Example 1 — Reverse percent and percent change

After a 20% discount a bicycle costs \$256. Find the original price, and then the percent increase needed to go from \$256 back to the original price.

$$0.80 P = 256 \Rightarrow P = \frac{256}{0.80} = 320.$$

$$\text{Percent increase from 256 to 320: } \frac{320 - 256}{256} = \frac{64}{256} = 25\% \text{ (divide by the old value, 256).}$$

Example 2 — Chained changes and percent of a percent

A price rises 20% and then falls 20%. In a school, 60% of students are in a club and 30% of club members play a sport. Find the net price change and the percent of all students who are club members and play a sport.

Net factor: $1.20 \times 0.80 = 0.96$, so the price is down 4%.

Percent of a percent: $0.60 \times 0.30 = 0.18$, i.e. 18% of all students.

Example 3 — Percentage points and a mixture

A rate rises from 12% to 15%: that is 3 percentage points, or $\frac{3}{12} = 25\%$ in relative terms. How many percent acid results from mixing 4 L of 10% acid with 6 L of 30% acid?

Pure acid: $0.10(4) + 0.30(6) = 0.4 + 1.8 = 2.2$ L in 10 L of mixture, so the concentration is 22%.

6 · PRACTICE SET — 30 PROBLEMS · EASY → MEDIUM → HARD → SAT CHALLENGE

Each problem states its tool. Write your work in the space below. Multiple choice: one correct option. Student-produced response (SPR): enter an exact value (fraction or decimal).

1. EASY · NO CALCULATOR NEEDED
A jacket has a regular price of \$80. During a sale, the price is reduced by 35%. What is the sale price of the jacket?

(A) \$45
(B) \$52 ✓
(C) \$108
(D) \$28

Worked solution

1. The discount is $0.35(80) = 28$ dollars.
2. Sale price = $80 - 28 = 52$. (Equivalently, $0.65 \times 80 = 52$.)

Answer: (B)

2. EASY · CALCULATOR
A random sample of 80 residents of a town was surveyed, and 18 of them said they commute to work by bicycle. The town has 12,000 residents. Based on the sample, what is the best estimate of the number of residents of the town who commute by bicycle?

Worked solution

1. Sample proportion: $\frac{18}{80} = 0.225$.
2. Apply it to the town: $0.225 \times 12,000 = 2700$.

Answer: 2700

3. EASY · CALCULATOR
The population of a town increased from 24,000 to 27,600. By what percent did the population increase?

(A) 115%
(B) 13%
(C) 15% ✓
(D) 87%

Worked solution

1. Change in population: $27,600 - 24,000 = 3600$.
2. Divide by the *old* value: $\frac{3600}{24,000} = 0.15 = 15\%$.

Answer: (C)

4. EASY · CALCULATOR
Maya answered 27 of the 60 questions on a practice quiz correctly. What percent of the questions did she answer correctly? (Enter the number without the percent sign.)

Worked solution

1. $\frac{27}{60} = 0.45$.
2. $0.45 = 45\%$.

Answer: 45

5.

EASY · NO CALCULATOR NEEDED

The table shows the study location preferred by a group of students, by grade. Of the students who prefer to study at home, what percent are in Grade 12?

	Library	Home	Café
Grade 11	30	42	18
Grade 12	20	28	32

(A) 40% ✓

(B) 60%

(C) 16%

(D) 35%

Worked solution

1. The group after “of” is the students who prefer home: $42 + 28 = 70$.

2. Of those, Grade 12 students: 28.

3. $\frac{28}{70} = 0.40 = 40\%$.

Answer: (A)

6.

MEDIUM · NO CALCULATOR NEEDED

The price of a stock increased by 40% in January. In February, the new price decreased by 25%. Over the two months, the price changed by which of the following?

(A) a 15% increase

(B) a 10% increase

(C) a 5% decrease

(D) a 5% increase ✓

Worked solution

1. January factor: 1.40. February factor: $1 - 0.25 = 0.75$.

2. Net factor: $1.40 \times 0.75 = 1.05$.

3. 1.05 means a 5% increase.

Answer: (D)

7.

MEDIUM · CALCULATOR

A small business's profit this year was 15% greater than its profit last year. This year's profit was \$46,000. What was last year's profit, in dollars?

Worked solution

1. Let L be last year's profit: $1.15L = 46,000$.

2. $L = \frac{46,000}{1.15} = 40,000$.

Answer: 40000

8.

MEDIUM · NO CALCULATOR NEEDED

In 2019, 8% of the adults in a certain town were unemployed. In 2023, 6% of the adults in the town were unemployed. By approximately what percent did the unemployment rate decrease from 2019 to 2023?

(A) 75%

(B) 2%

(C) 25% ✓

(D) 33%

Worked solution

1. The rate fell from 8 to 6, a change of 2 percentage points.

2. Percent decrease = $\frac{8 - 6}{8} = 0.25 = 25\%$ (divide by the old rate, 8).

Answer: (C)

9.

MEDIUM · NO CALCULATOR NEEDED

A store buys a lamp for \$60 and marks up the price by 50% to set the list price. During a sale, the lamp is sold for 20% off the list price. What is the store's profit on the lamp?

(A) \$72

(B) \$12 ✓

(C) \$18

(D) \$30

1. List price: $60(1.50) = 90$.**2.** Sale price: $90(0.80) = 72$.**3.** Profit = $72 - 60 = 12$ dollars.

10 Answer: (B)

MEDIUM · CALCULATOR

A survey of the students at a school recorded each student's grade and whether the student is in a club. The results are shown in the table. Of the students who are not in a club, approximately what percent are in Grade 11 or Grade 12?

	In a club	Not in a club
Grade 9	45	85
Grade 10	52	78
Grade 11	60	60
Grade 12	48	72

(A) 53%

(B) 55%

(C) 26%

(D) 45% ✓**Worked solution****1.** Students not in a club: $85 + 78 + 60 + 72 = 295$.**2.** Of these, Grades 11 and 12: $60 + 72 = 132$.**3.** $\frac{132}{295} \approx 0.447 \approx 45\%$.**Answer: (D)**

11.

MEDIUM · CALCULATOR

Positive numbers p and q satisfy $p = 1.3q$, so p is 30% greater than q . Of p , what percent is q , to the nearest whole percent? (Enter the number without the percent sign.)

Worked solution**1.** $q = \frac{p}{1.3}$, so $\frac{q}{p} = \frac{1}{1.3} \approx 0.769$.**2.** To the nearest percent, q is 77% of p (not 70%).**Answer: 77**

12.

MEDIUM · NO CALCULATOR NEEDED

A school has 2400 students. Of these, 45% ride the bus. Of the students who ride the bus, 40% live more than 5 miles from the school. How many bus riders live 5 miles or less from the school?

(A) 648 ✓

(B) 960

(C) 1440

(D) 432

Worked solution

1. Bus riders: $0.45(2400) = 1080$.

2. Riders living 5 miles or less away make up $100\% - 40\% = 60\%$ of riders.

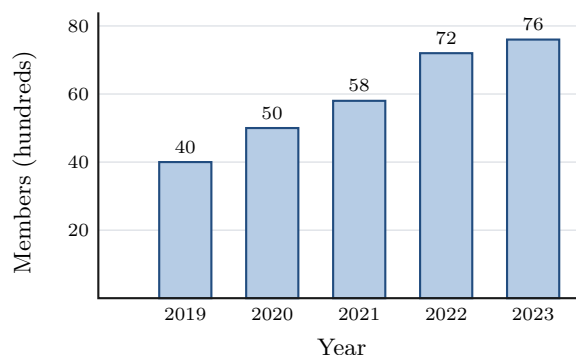
3. $0.60(1080) = 648$.

Answer: (A)

13.

MEDIUM · NO CALCULATOR NEEDED

The bar graph shows the number of members of a community center, in hundreds, at the start of each year from 2019 through 2023. Between which two consecutive years was the percent increase in membership greatest?



(A) 2022 to 2023

(B) 2019 to 2020 ✓

(C) 2020 to 2021

(D) 2021 to 2022

Worked solution

1. Percent increase = $\frac{\text{change}}{\text{old value}}$ for each pair.

2. 2019 to 2020: $\frac{10}{40} = 25\%$. 2020 to 2021: $\frac{8}{50} = 16\%$. 2021 to 2022: $\frac{14}{58} \approx 24.1\%$. 2022 to 2023: $\frac{4}{72} \approx 5.6\%$.

3. The greatest percent increase is from 2019 to 2020 (even though the largest numerical increase is 2021 to 2022).

Answer: (B)

14.

HARD · NO CALCULATOR NEEDED

The price of an item was increased by $x\%$, and then the new price was decreased by $x\%$, where $x > 0$. After both changes, the price was 96% of the original price. What is the value of x ?

Worked solution

1. Factors: $\left(1 + \frac{x}{100}\right) \left(1 - \frac{x}{100}\right) = 1 - \frac{x^2}{10,000}$.

2. $1 - \frac{x^2}{10,000} = 0.96 \Rightarrow x^2 = 400$.

3. Since $x > 0$, $x = 20$. (Check: $1.2 \times 0.8 = 0.96$.)

Answer: 20

15.

HARD · CALCULATOR

A customer bought a jacket after a 15% discount, and then used a coupon that took an additional 10% off the discounted price. The customer paid \$153. What was the original price of the jacket, in dollars?

Worked solution

1. Net factor: $0.85 \times 0.90 = 0.765$.

2. $0.765P = 153 \Rightarrow P = \frac{153}{0.765} = 200$.

Answer: 200

16.

HARD · NO CALCULATOR NEEDED

A positive number n is increased by 30%, and the result is then decreased by 30%. Consider the following statements.

I. The final value is 9% less than n .

II. The final value is 91% of n .

III. The final value equals n .

Which of the statements must be true?

(A) I only

(B) II only

(C) I and II only ✓

(D) I, II, and III

Worked solution

1. Final value: $n(1.30)(0.70) = 0.91n$.

2. I: $0.91n$ is 9% less than n — true. II: $0.91n$ is 91% of n — true. III: $0.91n \neq n$ for positive n — false.

3. Statements I and II only.

Answer: (C)

17.

HARD · NO CALCULATOR NEEDED

A customer buys n identical items that each have a list price of P dollars. A discount of $d\%$ is applied to the total list price, and then a sales tax of $t\%$ is applied to the discounted total. Which expression gives the total amount the customer pays, in dollars?

(A) $nP \left(1 - \frac{d}{100}\right) \left(1 + \frac{t}{100}\right)$ ✓

(B) $nP \left(1 - \frac{d}{100} + \frac{t}{100}\right)$

(C) $nP(1 - d)(1 + t)$

(D) $nP \left(1 + \frac{t}{100}\right) - \frac{d}{100}$

Worked solution

1. Total list price: nP .

2. After the discount, multiply by $1 - \frac{d}{100}$: $nP \left(1 - \frac{d}{100}\right)$.

3. Tax is applied to this discounted amount, so multiply by $1 + \frac{t}{100}$: $nP \left(1 - \frac{d}{100}\right) \left(1 + \frac{t}{100}\right)$.

4. The other options add the percents instead of chaining the factors, omit the division by 100, or subtract a percent as if it were dollars.

Answer: (A)

18.

HARD · NO CALCULATOR NEEDED

A 60-liter drink is 20% fruit juice by volume. How many liters of pure fruit juice (which is 100% juice) must be added to the drink so that the new mixture is 40% juice by volume?

Worked solution

1. Juice now: $0.20(60) = 12$ liters. Let x liters be added.
2. New juice: $12 + x$; new volume: $60 + x$. So $12 + x = 0.40(60 + x)$.
3. $12 + x = 24 + 0.4x \Rightarrow 0.6x = 12 \Rightarrow x = 20$.

Answer: 20

19.

HARD · CALCULATOR

A chemist mixes a 10% saline solution with a 50% saline solution to make 20 liters of a 35% saline solution. How many liters of the 10% solution are used?

(A) 10

(B) 12.5

(C) 5

(D) 7.5 ✓**Worked solution**

1. Let a be liters of 10% solution; then $20 - a$ liters of 50% solution are used.
2. Salt: $0.10a + 0.50(20 - a) = 0.35(20)$.
3. $10 - 0.4a = 7 \Rightarrow a = 7.5$ liters. (The 50% solution is 12.5 liters.)

Answer: (D)

20.

HARD · CALCULATOR

In a 2020 survey of 1200 randomly selected adults, 25% said they own an electric vehicle. In a 2023 survey of 1500 randomly selected adults, the percent who said they own an electric vehicle was 8 percentage points greater than in 2020. How many more adults in the 2023 survey than in the 2020 survey said they own an electric vehicle?

Worked solution

1. 2020: $0.25(1200) = 300$ adults.
2. 2023 percent: $25 + 8 = 33$, so $0.33(1500) = 495$ adults. (Percentage points add; they are not a 8% increase of 25%.)
3. Difference: $495 - 300 = 195$.

Answer: 195

21.

HARD · NO CALCULATOR NEEDED

The table shows the number of employees in each of three departments of a company and the percent of each department's employees who work remotely. Which of the following is closest to the percent of all employees in the three departments who work remotely?

Department	Employees	Percent working remotely
Design	120	25%
Sales	80	40%
Support	200	15%

(A) 30%

(B) 18%

(C) 23% ✓

(D) 27%

Worked solution

1. Remote employees: Design $0.25(120) = 30$; Sales $0.40(80) = 32$; Support $0.15(200) = 30$. Total 92.

2. All employees: $120 + 80 + 200 = 400$.

3. $\frac{92}{400} = 0.23 = 23\%$. (Averaging 25, 40, 15 gives about 27%, which ignores department sizes.)

Answer: (C)

22.

HARD · CALCULATOR

Coupon A takes \$15 off any purchase of \$60 or more. Coupon B takes 20% off any purchase. For how many whole-dollar purchase amounts P with $60 \leq P \leq 200$ is the price after using Coupon A strictly less than the price after using Coupon B?

Worked solution

1. Price with A: $P - 15$. Price with B: $0.80P$.

2. $P - 15 < 0.80P \Rightarrow 0.20P < 15 \Rightarrow P < 75$.

3. Whole-dollar values $60 \leq P \leq 74$: $74 - 60 + 1 = 15$ amounts. ($P = 75$ gives equal prices, which is not strictly less.)

Answer: 15

23.

HARD · CALCULATOR

The table shows the number of siblings reported by each student in a random sample of students from a school with 18,000 students.

Siblings	0	1	2	3 or more
Students	34	58	40	18

Based on the sample, which of the following is the best estimate of the number of students in the school who have fewer than 2 siblings?

(A) 6960

(B) 11,040 ✓

(C) 15,840

(D) 4080

Worked solution

1. Sample size: $34 + 58 + 40 + 18 = 150$.

2. Fewer than 2 siblings means 0 or 1: $34 + 58 = 92$ students.

3. Sample proportion $\frac{92}{150}$; estimate $\frac{92}{150} \times 18,000 = 11,040$.

Answer: (B)

24.

SAT CHALLENGE · NO CALCULATOR NEEDED

A store sets the list price of a jacket by marking up its wholesale cost by k percent. During a sale, the jacket is sold at 20% off the list price, and the store's profit is exactly 8% of the wholesale cost. What is the value of k ?

Worked solution

1. Let the wholesale cost be W . List price: $W \left(1 + \frac{k}{100}\right)$.
2. Sale price: $0.80W \left(1 + \frac{k}{100}\right)$. Profit = sale price $- W = 0.08W$, so the sale price is $1.08W$.
3. $0.80 \left(1 + \frac{k}{100}\right) = 1.08 \Rightarrow 1 + \frac{k}{100} = 1.35$.
4. $k = 35$.

Answer: 35

25.

SAT CHALLENGE · CALCULATOR

The price of an item was decreased by $x\%$, and then the reduced price was increased by $2x\%$. After both changes, the price was 12% greater than the original price. What is the sum of all possible values of x ?

Worked solution

1. Factors: $\left(1 - \frac{x}{100}\right) \left(1 + \frac{2x}{100}\right) = 1.12$. Let $u = \frac{x}{100}$: $(1 - u)(1 + 2u) = 1.12$.
2. $1 + u - 2u^2 = 1.12 \Rightarrow 2u^2 - u + 0.12 = 0$, so $u = \frac{1 \pm \sqrt{1 - 4(2)(0.12)}}{4} = \frac{1 \pm \sqrt{0.04}}{4} = \frac{1 \pm 0.2}{4}$.
3. $u = 0.3$ or $u = 0.2$, so $x = 30$ or $x = 20$. (Check: $0.7 \times 1.6 = 1.12$ and $0.8 \times 1.4 = 1.12$.)
4. Sum of all possible values: $20 + 30 = 50$. (Also by Vieta: $u_1 + u_2 = \frac{1}{2}$, so $x_1 + x_2 = 50$.)

Answer: 50

26.

SAT CHALLENGE · NO CALCULATOR NEEDED

From 2015 to 2020, the number of men in a club increased by 10%, the number of women in the club increased by 40%, and the total membership increased by 22%. What percent of the club's members in 2015 were women? (Enter the number without the percent sign.)

Worked solution

1. Let the 2015 membership be M with w women, so $M - w$ men.
2. 2020 total: $1.10(M - w) + 1.40w = 1.22M$.
3. $1.10M + 0.30w = 1.22M \Rightarrow 0.30w = 0.12M \Rightarrow w = 0.40M$.
4. Women were 40% of the 2015 members.

Answer: 40

27.

SAT CHALLENGE · NO CALCULATOR NEEDED

Quantity P is 140% greater than quantity Q , and quantity Q is 25% less than quantity R , where Q and R are positive. P is what percent of R ?

(A) 215%

(B) 105%

(C) 115%

(D) 180% ✓**Worked solution**

1. "140% greater than Q ": $P = Q(1 + 1.40) = 2.4Q$.
2. "25% less than R ": $Q = 0.75R$.
3. $P = 2.4(0.75R) = 1.8R$, which is 180% of R .

Answer: (D)

28.

SAT CHALLENGE · NO CALCULATOR NEEDED

A tank holds 80 liters of a cleaning solution that is 25% chemical by volume. First, x liters of water are removed by evaporation (the chemical does not evaporate). Then 10 liters of the pure chemical are added to the tank. The resulting solution is 60% chemical by volume. What is the value of x ?

(A) 40 ✓

(B) 50

(C) 20

(D) 30

1. Chemical at the start: $0.25(80) = 20$ liters. Evaporation does not change it.
2. After adding 10 liters of chemical: $20 + 10 = 30$ liters of chemical.
3. volume at the end: $80 - x + 10 = 90 - x$.

$$\frac{30}{90-x} = 0.60 \Rightarrow 90 - x = 50 \Rightarrow x = 40.$$

Answer: (A)

SAT CHALLENGE · CALCULATOR

The graph shows the percent of a company's total sales that came from each of three product categories, A, B, and C, in 2021 and in 2022. The company's total sales were \$2.0 million in 2021 and \$2.4 million in 2022. By what percent did the sales of Category B increase from 2021 to 2022?



(A) 20%

(B) 44% ✓

(C) 144%

(D) 5%

Worked solution

1. 2021 sales of B: $0.25 \times \$2.0 \text{ million} = \0.50 million .
2. 2022 sales of B: $0.30 \times \$2.4 \text{ million} = \0.72 million .
3. Percent increase: $\frac{0.72 - 0.50}{0.50} = \frac{0.22}{0.50} = 0.44 = 44\%$.
4. The share rose only 5 percentage points (20% in relative terms), but the total also grew, so the dollars grew by 44%.

Answer: (B)

30.

SAT CHALLENGE · NO CALCULATOR NEEDED

A survey asked n people whether they favor a proposal, and 40% of them said yes. Then 20 additional people were surveyed, and 15 of those 20 said yes. After these 20 people were included, exactly 45% of all the people surveyed favored the proposal. Finally, 10 more people were surveyed, all of whom said no. What percent of all the people surveyed now favor the proposal? (Enter the number without the percent sign.)

Worked solution

1. Yes answers after the 20 additional people: $0.40n + 15$. Total surveyed: $n + 20$.
2. $0.40n + 15 = 0.45(n + 20) \Rightarrow 0.40n + 15 = 0.45n + 9 \Rightarrow 0.05n = 6 \Rightarrow n = 120$.
3. Yes answers: $0.40(120) + 15 = 63$ (check: 63 of 140 is 45%).
4. After 10 more people (all no): 63 yes out of $140 + 10 = 150$, so $\frac{63}{150} = 0.42 = 42\%$.

Answer: 42